Adaptive Fraud Detection: Camouflage-Aware Graph Reconstruction under Concept Drift with Label-Scarce Retraining

Group 05 · Lin Yushan Edison, Melody Goh Xueni, Rohan Jaggi, Wong Yeow Kang

1. Problem Statement

Fraud detection is adversarial: detectors degrade because fraudsters observe what gets caught and adapt. GNNs rely on homophily, where connected nodes are alike. Fraudsters respond with camouflage, deliberately linking to legitimate accounts so neighbourhood averaging dilutes their signal toward the legitimate mean, and the aggregation mechanism itself conceals the fraud. Detecting this degradation conventionally requires recent labels, which arrive late and cost money, so drift is confirmed only after losses accrue.

Our claim: if camouflage attacks graph structure, the defence should **repair the structure** rather than patch the classifier. We use LLM-derived text evidence to **reconstruct the graph**, reweighting or removing edges the text does not support. Because this rewiring is label-free, it doubles as a monitoring signal as rising camouflage shifts the refined graph's homophily distribution, giving early warning of adversarial adaptation before ground truth arrives.

2. Data

YelpChi (~45,954 review nodes, ~14.5% fraudulent, 32 behavioural features, three relation types, dates supporting temporal splits), joined to raw review text.

3. Methodology

(a) Graph reconstruction: We first diagnose a per-node homophily ratio by relation, establishing that fraud nodes are less homophilous, then score each edge with an LLM comparing the two reviews' text for relative fit (specificity, experiential evidence, generic praise, stylistic distance), returning structured fields rather than a verdict. Scores become edge weights (soft) or thresholded removals (hard), both tested. The refined graph is scored once and reused: GraphSAGE trains on raw and refined graphs to isolate reconstruction from architecture, with CARE-GNN as the published camouflage-aware comparator. CatBoost calibrates node features plus LLM fields.

(b) Drift detection: PSI and KS on feature distributions and on the refined graph's homophily distribution, which is an unlabelled proxy for camouflage intensity, against rolling AUC-PR decay as the labelled comparator. Convergence of refined toward raw homophily means the rewiring is losing discriminative power.

(c) Label-scarce response: On trigger, pseudo-labels from Louvain communities on the refined graph and autoencoder reconstruction error, refined by confidence-based self-training. Challengers are promoted only on beating the incumbent on held-out recent data.

(d) Adversarial validation: Inject LLM-generated camouflage at increasing volume across temporal windows; measure degradation, trigger latency and recovery, raw versus reconstructed.

4. Evaluation

AUC-PR and recall@k under imbalance; strictly temporal splits. Raw-versus-refined graph is the primary comparison, held constant across four architectures. Headline metrics: detection latency (windows until drift flagged) and recovery gap (loss before promotion). Five seeds with reported variance; edge-scoring cost reported once per graph.

5. Anticipated Limitations

Reconstruction is label-free, so scoring errors propagate downstream, we sweep the pruning threshold to quantify sensitivity. Text scoring may penalise sincere but unspecific reviewers, examined via manual review of top false positives. YelpChi's natural drift may be insufficient, making synthetic injection load-bearing.